

Cambridge International AS & A Level

INFORMATION TECHNOLOGY**9626/13**

Paper 1 Theory

May/June 2026**MARK SCHEME**Maximum Mark: 70

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **17** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning	Use
	Correct point	Point-marked questions only: Section A, Section B part (a)
	Incorrect	Point-marked questions only: Section A, Section B part (a)
	Level 4	Levels-marked questions only: Section B part (c)
	Level 3	Levels-marked questions only: Section B parts (b) and (c)
	Level 2	Levels-marked questions only: Section B parts (b) and (c)
	Level 1	Levels-marked questions only: Section B parts (b) and (c)
	Level 0 – No creditable response	Levels-marked questions only: Section B parts (b) and (c)

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Annotation	Meaning	Use
Highlighter	Creditworthy part of an extended response	Levels-marked questions only: Section B parts (b) and (c)
EVAL	Evaluative point	Levels-marked questions only: Section B part (c)
	Omission or further development/ detail needed to gain credit	All questions
	Unclear or validity is doubted	All questions
DEV	Developed point	All questions
EG	Appropriate example or case study given	All questions
IRRL	Irrelevant	All questions
NAQ	Material that does not answer the question	All questions
	Highlighting a significant part of an extended response – to be used with another annotation e.g. IRRL or EVAL	Levels-marked questions only: Section B parts (b) and (c)
SEEN	1. Diagram or essay plan has been seen but no specific credit given 2. Additional page has been checked	1. Any diagrams or essay plans 2. All blank pages in the provided generic answer booklet and/or extension answer booklet(s).
R	Rubric error	Optional questions only (place at start of question not being credited): Section B (Candidates answer one question)

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Question	Answer	Marks	Guidance
1	THREE marks available: TWO from: • Data consists of raw facts and figures which require processing for interpretation (1) • Data does not have meaning (1) • Information is data that has a context (1) Allow converse for the above ONE mark for: • Use of ONE example to describe BOTH data and information (clearly stated as such) (1)	3	
2	THREE from: • ANY item of information if clearly identified as a password /his password (1) • ANY item of information if clearly identified as the answer to a security question (1) • Name of the bank account owner (1) • Address of the bank account owner (1) • Date of birth of the bank account owner (1) • Personal Identification Number (PIN) (1) • Bank account number (1)	3	

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Question	Answer	Marks	Guidance
3	FOUR from: Identify a benefit and state why it is a benefit OR cause of benefit Benefit e.g. • Data is safer/protected/secure (1st) • Stolen data cannot be understood (1st) • Reduced desire to hack (1st) • Data is unusable if stolen/accessed (1st) Explanation e.g. Any suitable explanation	4	
4(a)	TWO marks available: • High quality data is (more) accurate/reliable/detailed/relevant/up-to-date/complete//data needs to be/should be accurate (1) • and decisions based on them are more likely to be correct//decreases risk of making an incorrect decision (1) • provides better insights into shopping habits (1) Answer may be focused on one factor: e.g. • Data must be up-to-date (1) because trends change over time /you know what is popular now (1) ACCEPT CONVERSE of these arguments where applicable	2	

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Question	Answer	Marks	Guidance
4(b)	EIGHT from: - The report holds information that may be useful, but there are reasons why it may not be (1) Three marks may be awarded where relevant issues are identified, but not explained, where no other marks would be awarded overall other than Mark Point 1 (above). Benefits (Max six) e.g. - It is data that has been collected for the similar purpose (1) - The data already exists (1) so there is no time required to gather the data (1) decision can be made more quickly (1) - The data has already been organised (1) therefore it is not raw data, so no organisation required (1) - Data will be complete/lots of detail (1) - It is from a larger sample (1) removes bias (1) - Allows access to data that may be (currently) impossible/difficult to gather (1) COSTS - The costs of gathering the data (to the shop owner) are zero/minimal/reduced (1) - Identified saving cost of gathering data (1)	8	Answers must be linked to using the data to identify shopping habits. Do Not Award - Data is untrustworthy - Data is up-to-date

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Question	Answer	Marks	Guidance
4(b)	Drawbacks (Max six) • The data sample is national (1st) therefore not focussed enough//irrelevant (1) • The data is old// is now out of date (1) so trends//shopping habits may have changed (1) • Report may be biased (NOT 'false') (1) Conclusion One mark for a value judgement/conclusion based on two arguments (positive or negative or both) (1)		

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Question	Answer	Marks	Guidance
5	SIX from: MAX 5 if no reference to weather forecasting Benefits (Max four) • Very fast processing//has a high FLOPS (1) • Weather forecast can be produced quickly (1) • Can handle a great deal of information (NOT store) (1) • Produce more/very detailed forecasts (1) • Consider a wide range of possible scenarios (1) • Decision is data driven (1) Drawbacks (Max four) • Supercomputers struggle to predict extreme events//working with extreme data (1) • Lacks human intuition (1) • Supercomputers have a limited lifespan (1) • Supercomputers are expensive to buy (1) • Supercomputers require specialist maintenance teams (1) • Supercomputers are very large//take up a lot of space (1) • Supercomputers require a lot of heat management//generates a lot of heat (1)	6	Do Not Accept: • ...and supercomputers can do this Some marks are awarded based on the use of <u>superlatives</u>. Candidates MUST have a clear awareness of the absolute vastness of the processing speed, or of the amount of information able to be handled. Therefore, if this is established, words such as 'a lot' or 'a great deal' can be awarded.

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Question	Answer	Marks	Guidance
6(a)	ONE from: - The rearrangement/reorganisation of data/files (on a storage medium) to ONE OF (1) - create contiguous files - ensure efficient storage/access	1	Do Not Award Move or put together
6(b)	ONE from: - Modern computers use SSD storage (1) - SSD storage does not spin/is static (1) - SSD storage does not fragment files (1) - Defragmentation of an SSD reduces its (effective) life (1)	1	

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Question	Answer	Marks	Guidance
6(c)	FOUR marks available (TWO per fully described type of utility software): e.g. • Anti-Virus (1st) Detect/remove viruses/malware (1) • Data compression (1st) To create extra storage/decrease the file size of files (1) • File copying (1st) To make another/ a second version of a file (1) (do not accept 'to copy the file') • Disk formatting (1st) To prepare a disk for first use//to return to disk to factory settings (1) Verify all tracks/sectors are ready for recording data (1) • Disk cleaners (1st) Remove unwanted files from the hard disk (1) • System optimisers (1st) Manage system resources so that computer's performance is improved (1) • Disk encryption (1st) Converts data on the disk into an unusable format (1) • Firewall (1st) Monitors/controls network traffic (1) Acts as a barrier between the network and the outside world (1)	4	Do Not Accept: • Back-up//recovery • File deletion • Disk defragmentation

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Question	Answer	Marks	Guidance
7	EIGHT from: - Because there is a digital divide (between them) (1) Possible themes include: - Different locations (1) with different access (1) - Lower disposable income/affordability/socio-economic difference (1) affecting access (1) - Different levels of fear (of any type)/comfort (1) however, this fear will quickly disappear with use (1) if issue is not addressed, this problem will get worse (1) - Different day to day experience/training (1) - Different jobs make different use of IT (1) - Disability (1) - Lack of awareness of possibilities (1) - Any example of choice or lifestyle (absence of or different need lack of interest, lack of online friends (1) Answers may be awarded for: Identification of issue (1) Explanation/description of issue (NOT just another version of the same issue) (1)	8	

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Question	Answer	Marks	Guidance
8(a)	TWO from: - In bold (1) Centrally aligned//centred horizontally/vertically (1)	2	Do Not Accept Border – not a content
8(b)	THREE marks available: Normal test data (max ONE) - Data that should produce expected results (1) - Data that is within acceptable boundaries//correct range (1) - Should be accepted by the cells (1) - Sensible data (1) Extreme test data (max ONE) - Data that is at the boundary (1) - At the limit of acceptability (1) Abnormal test data (max ONE) - Falls outside acceptable limit/range (1) - Data that is invalid (1)	3	
8(c)	THREE marks available, one per type of test: Normal - Any single number between 44 and 50 (inclusive) (1) Extreme 44//50 (ONE ONLY – NOT BOTH) (1) Abnormal - Any other single number beyond range 44 - 50 (1) - Any data other than numeric (1)	3	Accept real values between 44 and 50 Answer must be actual data (e.g. 50) and not description of data (e.g. 'a value between 44 and 50'). If descriptions given, do not accept this as non-numeric for abnormal test data

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Question	Answer	Marks	Guidance
8(d)	FIVE from: $=\text{IF}(\text{F3}>\text{C3},\text{"Check Score"},(\text{IF}(\text{F3}>\text{B3},\text{A3},(\text{IF}(\text{F3}>\text{B4},\text{A4},(\text{IF}(\text{F3}>\text{B5},\text{A5},\text{A6}))))))$ - The formula will not produce the correct grade for all scores (1) at the (lower) boundary of each grade (1) it will produce the grade below (1) Wrong inequality symbol marks (route 1) - The formula only uses an inequality symbol (1) and should have used an inequality (accept equivalent statement, such as > has been used) AND an equals statement (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) Wrong number marks (route 2) - The wrong value has been used as the lower boundary (1) - The formula uses a greater than inequality (1) the value from the top of the lower boundary should be used (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) One mark may be awarded for: - One mark for showing how the formula may be improved:	5	Marks may be awarded from route 1 OR route 2 only. Where candidate mixes both explanations, mark to the candidate's best advantage.

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Question	Answer	Marks	Guidance
8(d)	e.g. • IF(F3 > B3 – 1 • IF(F2 = B3 etc. IF(F2 > 43		

Question	Answer	Marks	Guidance
9(a)	ONE mark available: • A database that stores data in more than one linked table (1) • It has linked tables (1)	1	Do Not award • Entities • Multiple databases • Multiple locations • Answers such as ‘two tables with relationships’ Use of relationship here – mark as REP
9(b)(i)	TWO from: • Contains unique data (1) • Cannot contain a null value//must contain a value (1) • Can be based on one cell OR composite/based on two cells (1) (whole answer required) • Only one per table (1)	2	
9(b)(ii)	ONE from: • It enables links to be formed between tables (1) • It uniquely identifies a record (1) • It may be used in a different table as a foreign key (1) • It maintains accuracy/consistency of data (1) • It allows for quick access to individual records (1)	1	

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Question	Answer	Marks	Guidance
9(c)	ONE mark available: - A compound key is two or more pieces of data//a primary key (for an individual record) is a single piece of data (1) - A compound key is based on more than one column//a primary key is based on one column (1) - A compound key gives uniqueness from two columns of data (or more)//a primary key gives uniqueness from one column of data (1)	1	
9(d)	FOUR from: - All data must be atomic (1) - There must be a primary key (1) - There must be no duplicated rows/columns (1) - There must be only one value for each row of each column (1)	4	
9(e)	FOUR from: - Any change to one record which is needed can instantly be made to any related records/improved data consistency (1) - Provides referential integrity (1) - The database does not have redundant data (1) - Makes the file size small/less storage needed (1) - There is no/less data duplication/repeating data (1st) so there are fewer errors in the data (1) so there is less chance of storing incorrect copies of the data (1) - There is less data to modify (1) - Change (to data) can be made more quickly (1)	4	Do Not Award No repeat fields

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Question	Answer	Marks	Guidance
10(a)	TWO from: • Uses IF THEN reasoning (1) • Uses forward/backward chaining (1) • Applies the rules to the data//interrogates the knowledge base (1) • To identify possible solutions/answers/results (1) • Application to identification of flowers (must be more than mentioning flowers and must be more than a repeat of question) (1)	2	
10(b)	TWO from: • Holds all the facts (1) such as named item from identification of flowers (1) • Holds all the rules (1) • Holds all the heuristics (1)	2	